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COMP4139

Machine Learning (MLE)

- Feature Selection & Dimensionality Reduction

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Learning Objectives

- Understand the Feature Selection (FS) and Dimensionality Reduction (DR)
- Know when to use FS and DR
- Understand the working principle of key FS methods
- Understand the working principle of PCA & LDA
- Understand the concept of manifold learning



Aim of Feature Selection and Dimensionality Reduction

- Reduce the impact caused by *Curse of Dimensionality*
- Remove redundant features to improve performance
- Increase computational efficiency
- Reduce cost in new data acquisition
- FS vs DR
 - FS retain a subset of the original features
 - DR generate a new set of features that is compact but does **NOT** retain the original meaning of features.



Things to consider when using FS and DR

- The target dimension
- Interpretability
- Feature correlations/dependency
- Feature reliability and repeatability
- Methods (different methods likely to result in different features)



- Wrapper methods
 - Search for optimal feature subset that maximise the decision-making performance
 - Methods: recursive feature elimination; sequential feature selection.
- Embedded methods
 - Integrate the FS process to the model learning process.
 - Methods: ridge (ElasticNet); lasso; random forest (feature ranking).
- Filter-based methods
 - Selection is based on feature relationships and statistics rather than performance.
 - Methods: univariate (ANOVA); Chi Square; correlation/variance

Forward Feature Selection (Wrapper Method)

Let $F = \{F_1, \dots, F_n\}$ be the pool of potential features and let $M(X)$ be the evaluation metric for feature set X .

```
1   $X \leftarrow \emptyset$ 
2  while  $X \neq F$ 
3       $B \leftarrow 0$ 
4       $Y \leftarrow \emptyset$ 
5      for each  $X_i \in F \setminus X$ 
6          if  $M(X \cup \{X_i\}) > B$  then
7               $B \leftarrow M(X \cup \{X_i\})$ 
8               $Y \leftarrow X \cup \{X_i\}$ 
9      if  $M(X) > B$  then
10         return  $X$ 
11     else
12          $X \leftarrow Y$ 
13 return  $X$ 
```

Features that belongs to F not X

- . **X**: the final selected feature set
- . **B**: is stored best evaluation metric value
- . **Y**: the selected feature set at each iteration
- . **M**: The evaluation metric, e.g. Entropy; Classification rate; Regression error
- . **Recursive feature elimination** method is similar: starts with the full set and eliminate one at a time.



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LASSO (Embedded Method)

- LASSO (least absolute shrinkage and selection operator)
- Add a L1 regularisation term to reduce the number of effective features
- The loss function is not differentiable. Sub-gradient methods or least-angle regression can be used to optimise the loss

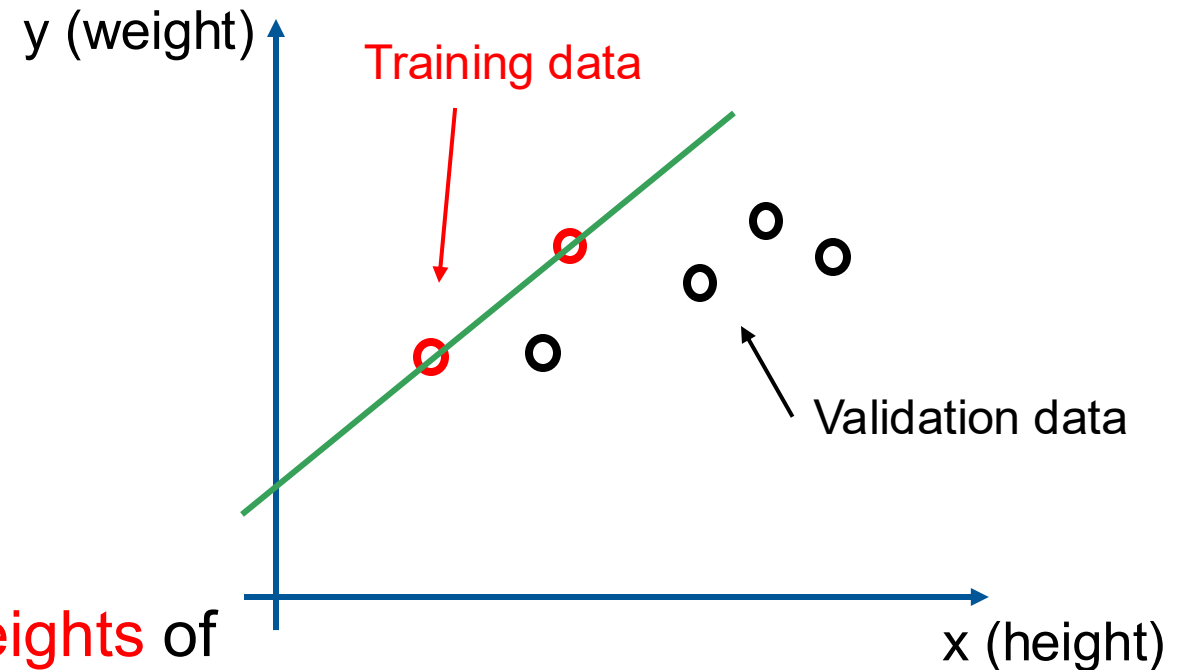
$$\min_w (\| \mathbf{X}\mathbf{w} - \mathbf{y} \|^2 + \lambda |\mathbf{w}|)$$

Least squared term L1 regularisation term

For multiple features:

e.g. $y = w_0 + w_1 * x_1 + w_2 * x_2 + w_3 * x_3$

LASSO a **higher** λ will make some of the **weights** of x becomes **0**, hence reduce the dimensionalities.



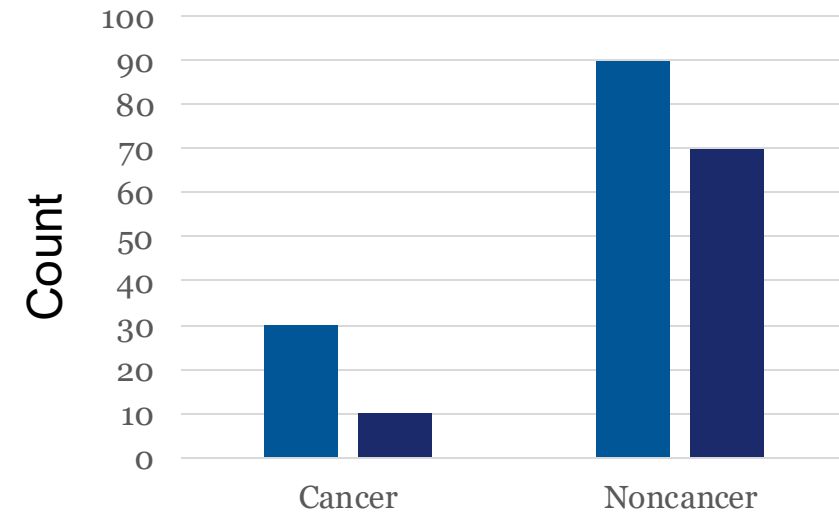


Chi Square vs T-test vs ANOVA (Filter Method)

- Univariate feature selection (assuming features are independent to each other)
- A chi-square tests the independence of predictor and outcome event, suitable for **categorical** features in **categorical** outcome.
- T-test compares the statistical difference of two groups (**binary class**) and used for **continuous** features
- ANOVA uses variance to test the relationship between **categorical** predictors and **continuous** outcome response (e.g. gender, age group to predict exam mark).
- Correlation test work for predictors and outcome are both **continuous**.
- Assume a **null-hypothesis**. Use p value to **reject** the null-hypothesis (e.g. **$p < 0.01$**). Smaller p value indicates the evidence against the null-hypothesis.

Chi Square

How categorical predictor associated with the categorical outcomes



$$X^2 = \sum \frac{(O - E)^2}{E}$$

Where:

- X^2 is the chi-square test statistic
- Σ is the summation operator (it means “take the sum of”)
- O is the observed frequency
- E is the expected frequency

null-hypothesis: the two categorical data are independent



Chi Square

How categorical predictor associated with the categorical outcomes

Gender	Cancer	No Cancer	Total
Male	30	90	120
Female	10	70	80
Total	40	160	200

Gender	Cancer (Expected)	No Cancer (Expected)
Male	$(120 \times 40) / 200 = 24$	$(120 \times 160) / 200 = 96$
Female	$(80 \times 40) / 200 = 16$	$(80 \times 160) / 200 = 64$

$$\chi^2 = \sum \frac{(O - E)^2}{E}$$

Gender	Cancer	No Cancer
Male	$(30 - 24)^2 / 24 = 1.5$	$(90 - 96)^2 / 96 = 0.375$
Female	$(10 - 16)^2 / 16 = 2.25$	$(70 - 64)^2 / 64 = 0.5625$

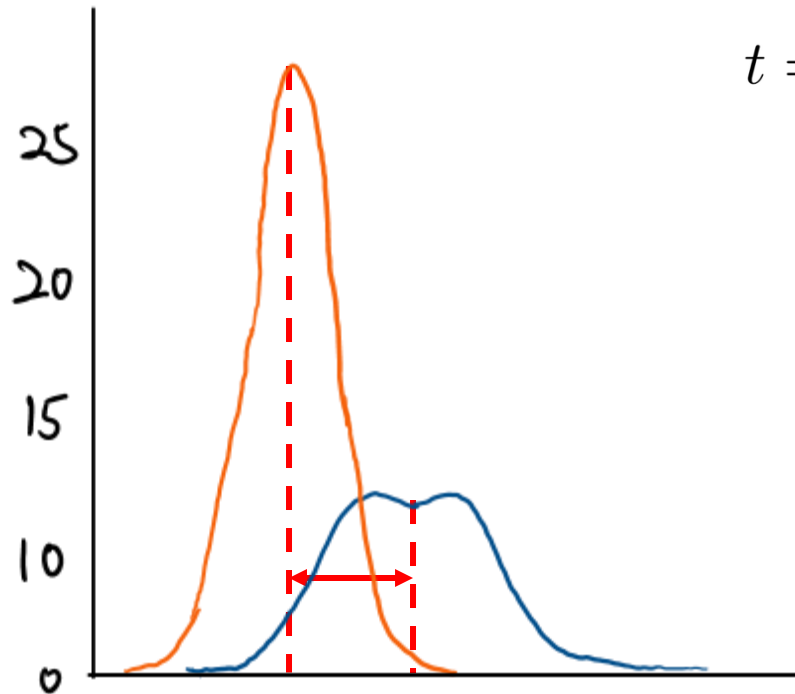
$$\text{Total } \chi^2 = 1.5 + 0.375 + 2.25 + 0.5625 = 4.6875$$

At $df = 1$, the critical χ^2 (0.05) is 3.841.

Since $4.6875 > 3.841$, we reject the null hypothesis — there is a significant association between gender and cancer incidence.

How the mean of two groups are different from each other

Number of occurrence



$$t = \frac{\bar{X}_1 - \bar{X}_2}{s_p \sqrt{\frac{1}{n_1} + \frac{1}{n_2}}}$$

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

$\bar{X}_1, \bar{X}_2$ = sample means of the two groups

s_1, s_2 = sample standard deviations of the two groups

n_1, n_2 = sample sizes of the two groups

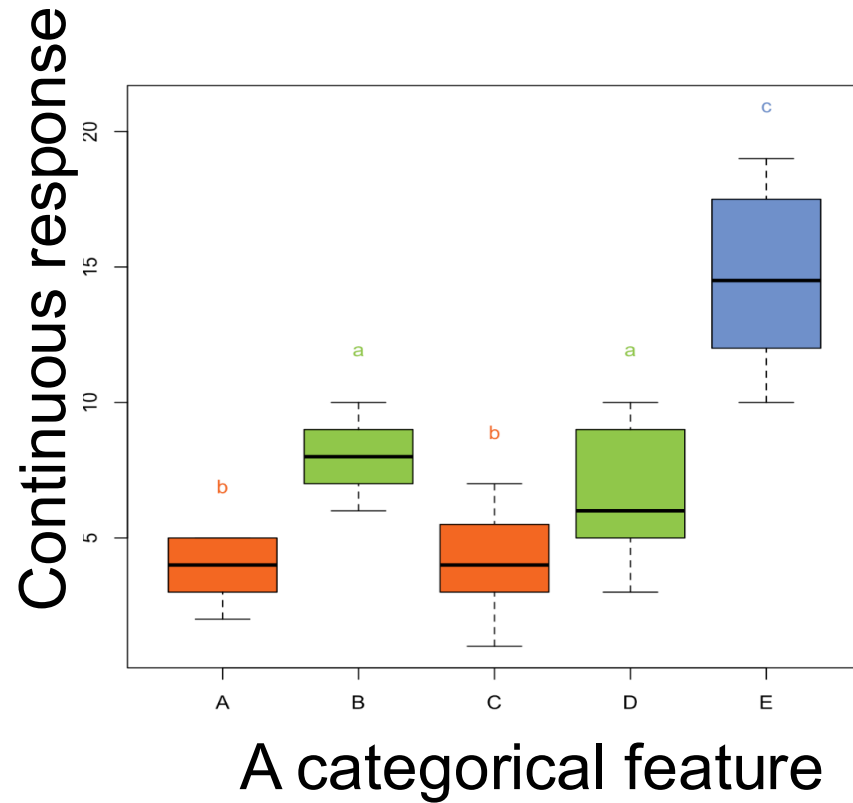
s_p = pooled standard deviation

null-hypothesis: the mean of the two groups are the same



ANOVA

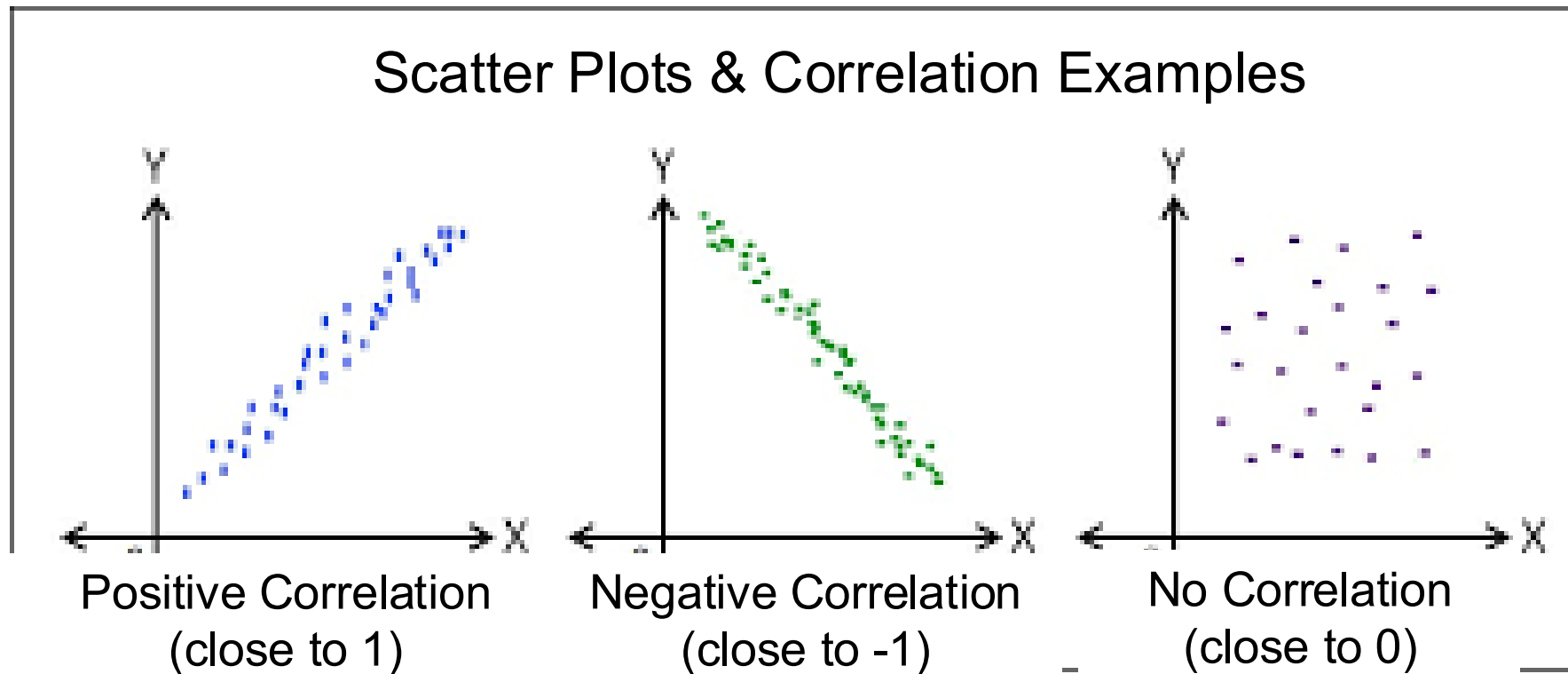
ANOVA checks variance between groups of a categorical feature with respect to continuous output. Variance between the groups and variance within the groups.



null-hypothesis: the variance across categories are equal.

Correlation

Correlation of continuous predictor and continuous outcome (e.g. Pearson correlation coefficient)



null-hypothesis: the two group of data are not correlated.



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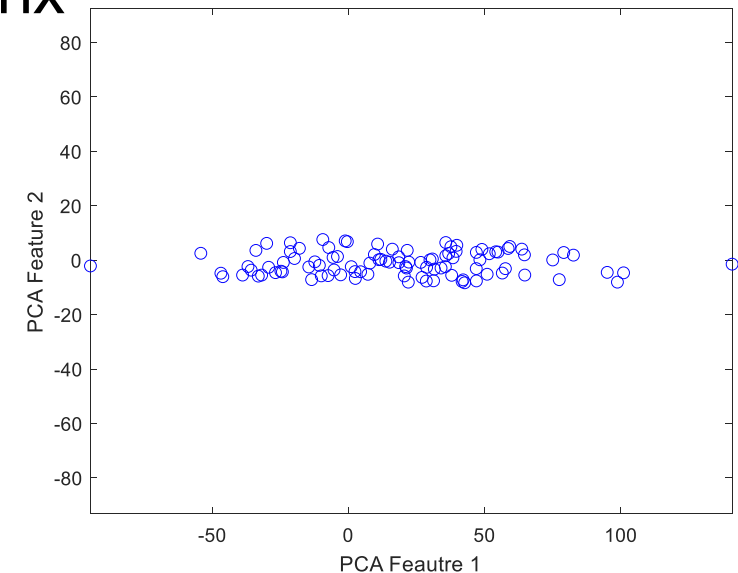
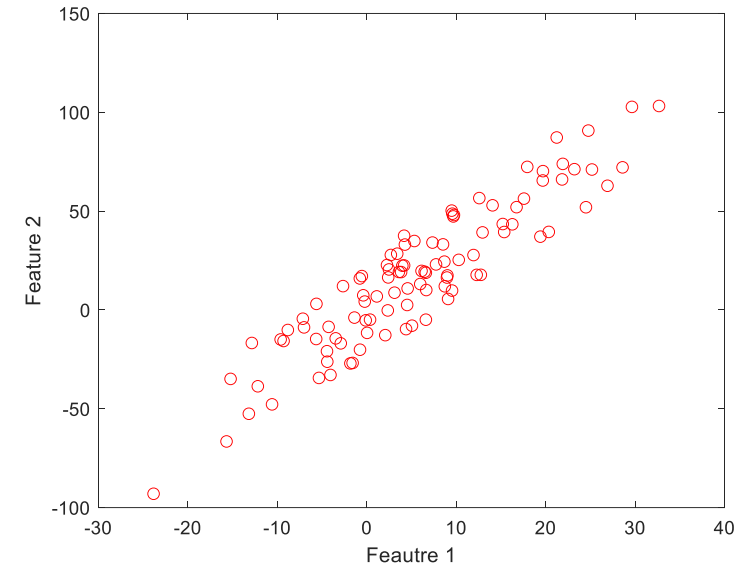
Popular Dimensionality Reduction Methods

- Principal Component Analysis (PCA)
- Linear Discriminant Analysis (LDA)
- Manifold learning (non-linear)



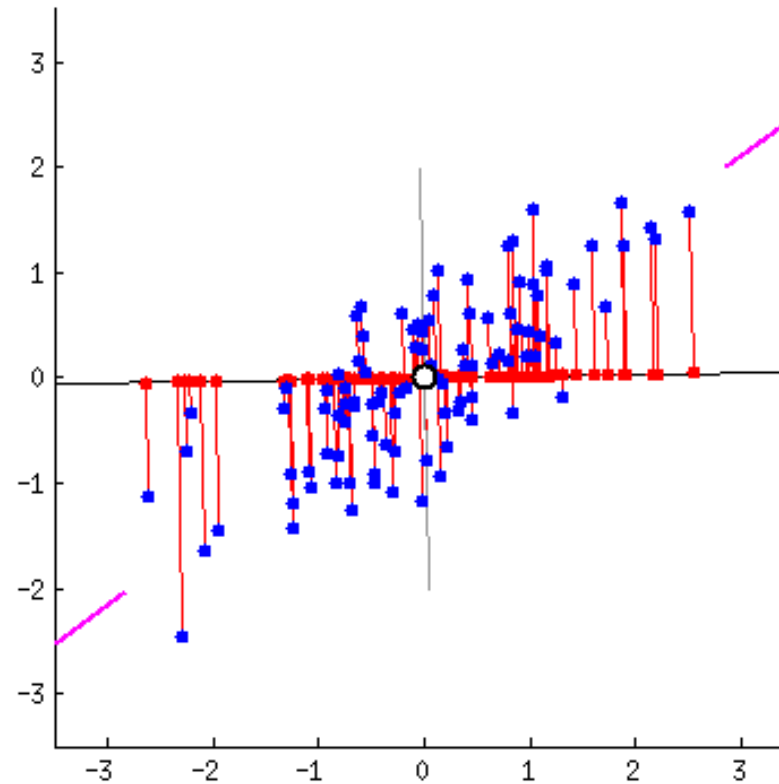
Principal Component Analysis

- Two methods resulted the same PCA calculation.
 - Maximum variance
 - Minimise average projection error
- PCA requires calculation of:
 - Mean of observed variables
 - Covariance of observed variables
 - Eigenvalue/eigenvector computation of covariance matrix
- How to calculate Eigen vectors and Eigen values:
<https://www.youtube.com/watch?v=TQvxWaQnrqI>





Principal Component Analysis



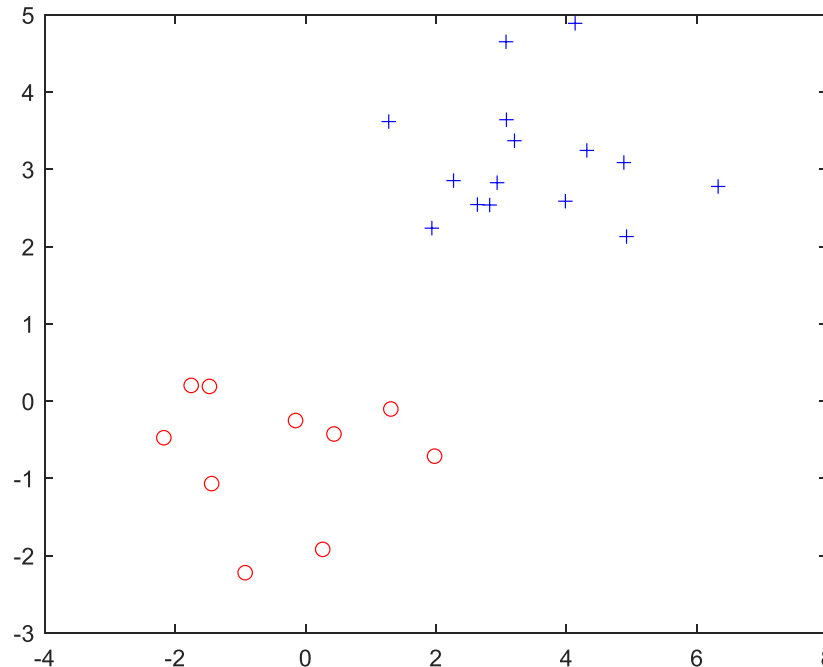
Linear Discriminant Analysis

- LDA is a predictive modelling algorithm for multi-class classification.
- Dimensionality reduction by providing a projection of a training dataset that best separates the examples by their assigned class.
- PCA :unsupervised; LDA: supervised
- How multi-class works?

$$J(w) = \frac{(m_1 - m_2)^2}{s_1^2 + s_2^2}$$

Class mean

Class variance



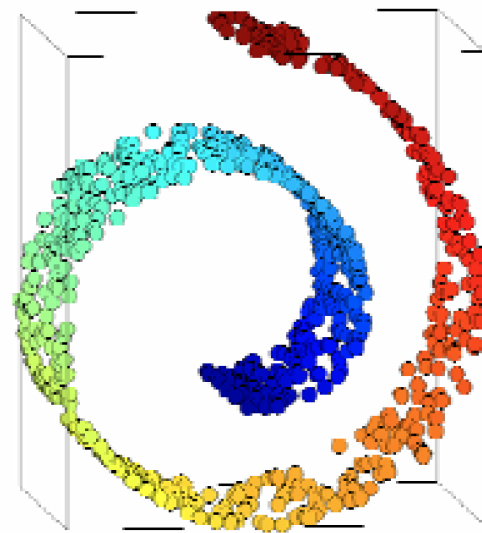
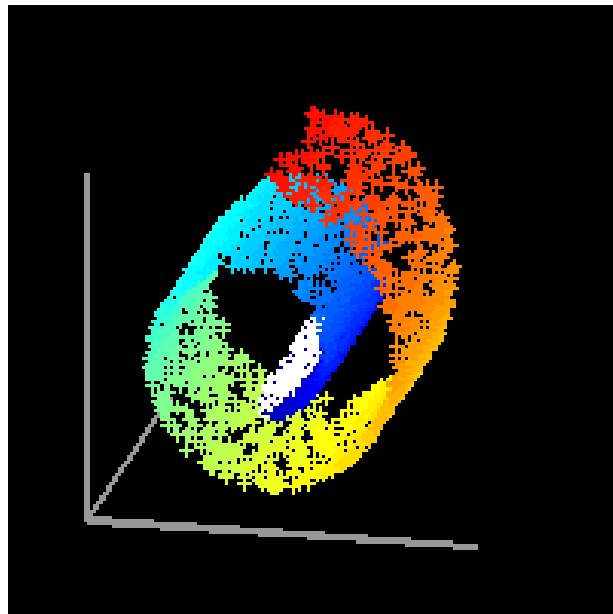
- The intuition: high-dimensional datasets often vary due to only a small number of parameters.
- E.g. dataset of 25, 64x64 images:



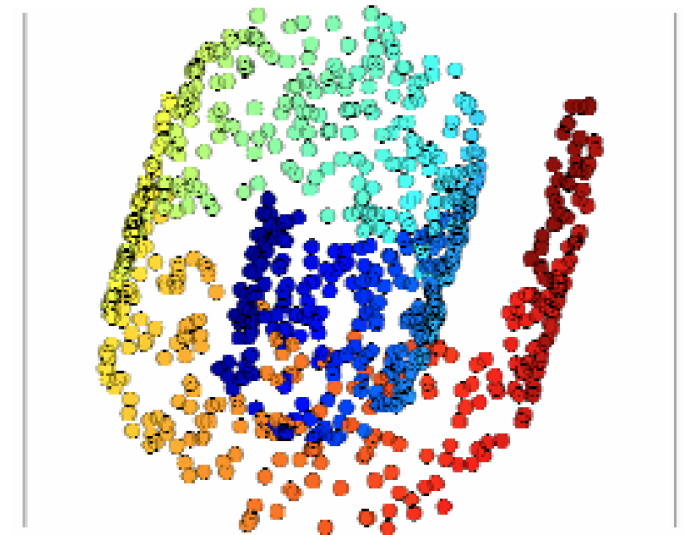
- Dataset actually varies due to only **parameters**: 2 angles + 1 illumination
- The images span a 3-dimensional **manifold** of R^{4096}

Manifold Learning

- Manifold learning aims to learn the latent representation of the original data in lower dimensions.
- PCA is a linear manifold learning method, which doesn't work in some cases.
- Swiss roll: what is the minimum dimension to present the data?



3D

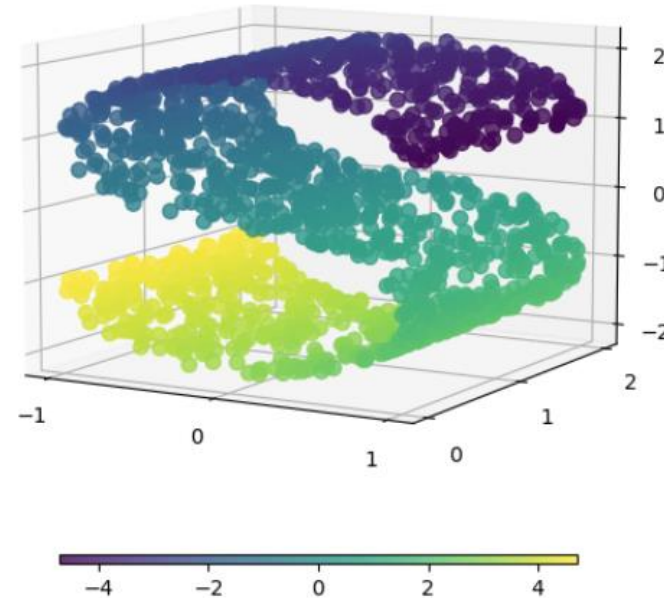


2D



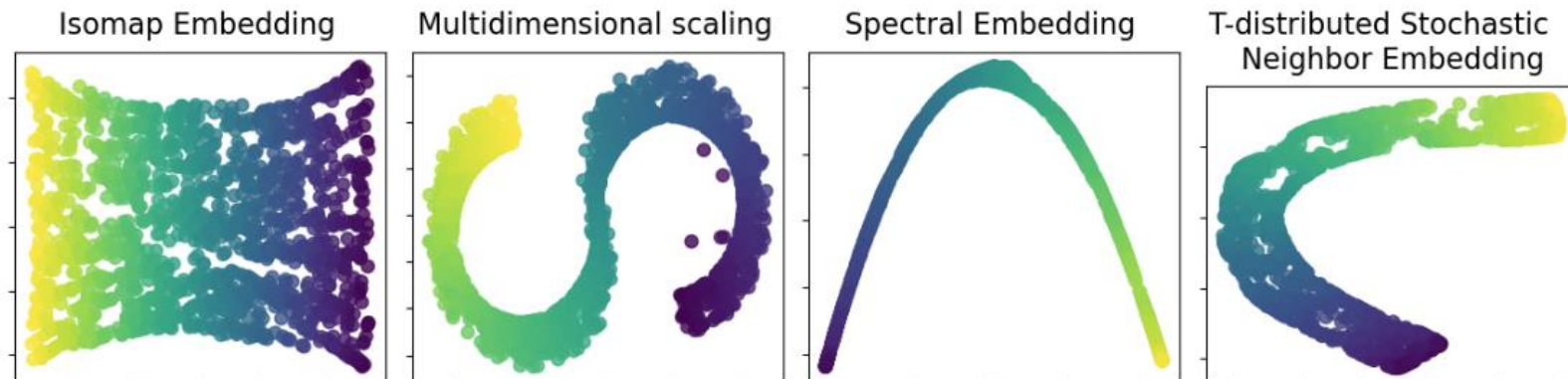
Nonlinear Manifold Learning Methods

Original data in 3D



Similar ideas to preserve local neighbouring relationships

Learned Manifold
in 2D





Key Messages

- Understand the difference of FS and DR
- Know when to FS and DR
- Understand the working principle of key FS methods
- Understand the working principle of PCA & LDA
- Understand the concept of manifold learning
- Know some of the non-linear manifold learning methods